

*3 Define forward and backward chaining

->forward-is a reasoning process that starts with known facts and applies rules to deduce new facts

- data driven approach

- starts from initial facts

- moves forward

backward-starts with a goal and works backward to find the facts that support it

- goal driven approach

- starts from goal state

- moves backward

*4 what are events

->is a set of outcomes of a random experiment

*6 what is NLP

->natural language processing

nlp subfield of AI that deals with interaction between computers and humans in natural language

nlp is a branch of ai that enables computers to understand and generate human lang in a meaningful way

it combines comp science to allow to process spoken and written lang for tasks like lang translation, sentiment analysis, speech recognition and question answering

4M

7 explain AO Algorithm with example

->A* with multiple goals

- is a search algo

- used to find the shortest path to multiple goals

- And-Or-star algo used in AI for solving problems

it is used for problem reduction, where complex problems are break down into smaller subproblems

- AO* represented as AND-OR graph

- AND NODE - require all child node to be solved to achieve the parent goal

- OR NODEs - require only one of the child nodes to be solved to achieve the

parent goal

heuristic function - guides the search by estimating the cost to reach the goal from a given node

- eg. buying a car

- consider the goal of buying a car represented in an AND-OR graph

- goal - G=buy a car

- option 1 (OR) - steal the car(low cost)

- option 2 (OR) - earn money AND buy the car

- earn money can be an AND node itself(eg get a job AND save money)

- the AO* algo evaluates the cost

*8 translate following English statement into FOL(first-order-logic)

->fol is a formal system for expressing and reasoning about logical stmts

- I)All birds fly

->ultra $\forall x(\text{Bird}(x) \Rightarrow \text{Flies}(x))$

- II)Every man respects his parent

->ultra $\forall x(\text{Man}(x) \Rightarrow \text{Respects}(x, \text{Parent}(x)))$

*9 write state space representation of monkey-banana-problem also specify initial final state and list operator or rules

->A classic program in AI problem, where a monkey in a room trying to reach a bananas hanging out of reach. A box is available to help

State Space Representation - a state can be represented as a tuple or structure

```
state(MonkeyHorizontalPos,MonkeyVerticalPos,BoxHorizontalPos,hasbananastatus)
```

```
monkeyHP:{atdoor,atwindow,middle,underbananas}
```

```
monkeyvp:{onfloor,onbox}
```

```
boxhp:{atdoor,atwindow,middle,underbananas}
```

```
hasbs:{hasnot,has}
```

```
Initial State- state(atdoor,onfloor,atwindow,hasnot)
```

```
final state- state(onbox,underbananas,has)the monkey has the banana
```

```
operators(rules)-move,climb,push
```

```
move(p1,p2) - monkey moves from position p1 to p2 on the floor
```

```
climb(p) - monkey climbs onto the box at position p
```

```
push(p1,p2) - monkey pushes the box from p1 to p2 ( monkey must be on the floor at p1)
```

*10 Write a python program to find factorial of a given number

```
def fact(n):  
    if n < 0 :  
        return "Invalid Number"  
    elif n==0 :  
        return 1  
    else :  
        f=1  
        for i in range(1,n+1) :  
            f=f*i  
        print("factorial of the num is = ",f)
```

```
n=5
```

```
fact(n)
```

*11 what is partially observable game? give its example

-> a game where the agent does not have complete knowledge/info of the game state

players can only observe a part of the game state and must make decisions based on this limited info

eg.poker,where players do not know each others hands

*12 explain minmax procedure with example

-> it is a decision making algo used in game theory to minimize the maximum potential loss

is a decision making algo used in game theory and AI for two player such as chess,tic tac toe

the procedure works recursively

max player - tries to maximize their score

min player - tries to minimize the max players score

procedure

1. generate the entire game tree down to the terminal state(game over)

2. assign scores to the terminal states

3. back the scores up the tree

4. select the moves at the current state

eg.tic tac toe

in tic tac toe ,terminal state are evaluated as +1 for a win,-1 for a loss,and 0 for a draw

the max player chooses moves that lead nodes with higher values

while the min player chooses moves leading to lower values

the algo used for the best possible outcome

*13 explain generate and test algorithm

-> is a simple algo that generates possible solutions and tests them to find the

correct one

is a simple problem solving method in AI

it is non informed search technique

algo steps

generate - generate possible solution

test - test possible solution

loop - if the solution is not found then repeat steps until solution found

eg.

to open a combination lock the algo generates every possible number sequence

tests each one on the lock until it opens

*15 what are the application of NLP

-> sentiment analysis - determine the emotional tone eg customer review

language translation

text summarization - automatically generate summaries of longer documents

chatbots

machine translation - automatically translating text from one lang to another

eg google translate

spam filtering - analysing email content to classify msgs as spam or not

smart assistants - powering virtual assistants

question answering systems

*16 give the state space representation of "water jug problem" where there are 2 jugs of 4L and 3L respectively we want 2L water in 4L jug

-> we have two jugs, a 4L(jug A), a 3L(jug B)

state space representation - (A,B)

where A is the amount of water in 4L

B is the amount of water in the 3L

initial state - (0,0)

both jugs are empty

final state - (2,)

operator(rules) - fill,pour,empty

fillA : (A,B) ->(4,B) (fill the 4L jug)

fillB : (A,B) ->(A,3) (fill the 3L jug)

emptyA: (A,B) ->(0,B) (empty the 4L jug)

emptyB: (A,B) ->(A,0) (empty the 3L jug)

pourAtoB

pourBtoA

*17 list the criteria to measure the performance of different search strategies

-> completeness - is the algo to find a solution if one exists

optimality - is the algo to find the best solution if multiple solution

exists

time complexity - how much time does the algo take to run

space complexity - how much memory does the algo require

*19 what is difference between artificial intelligence and natural intelligence

-> AI- refers to artificial intelligence

created by humans(machines,comp)

artificial,man-made in nature

high speed complex calculations

learns through datasets

rule based reasoning

NI- refers to natural intelligence which is the intelligence exhibited by humans and animals

created by nature

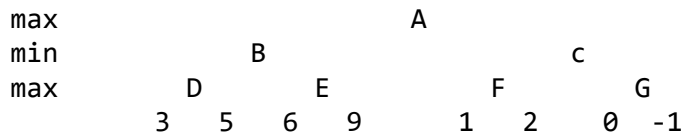
natural in nature

*20 write a python program display any four list operation

2. node A (max level) -

this is the root node (max level),
 which chooses the maximum value from its children (D,C,B)
 $A: \max(\text{value of D}, \text{value of C}, \text{value of B}) = \max(0, 2, 2) = 2$
 the minimax value for each node is as follows :
 node D : 0
 node C : 2
 node B : 2
 node A : 2
 the minimax value for the max node is 2

- Use the Minimax algorithm to compute the minimax value at each node for the game tree below



1. node D,B,F,E(max level):
 $D: \max(3, 5) = 5$
 $E: \max(6, 9) = 9$
 $F: \max(1, 2) = 2$
 $G: \max(0, -1) = 0$
2. node B,C(min level) :
 $B: \min(\text{value of D}, \text{value of E}) = 5$
 $C: \min(\text{value of F}, \text{value of G}) = 0$
3. node A (max level) =
 $A: \max(\text{value of B}, \text{value of C}) = 5$
 the minimax value for the each node as follows :
 node D: 5
 node E: 9
 node F: 2
 node G: 0
 node B: 5
 node C: 0
 node A: 5
 the minimax value for the max node is 5

NOTES

- What is equality?
 is the state of beign equail in particular status
 in mathematic and logic it refers to a relationship between two expressions that have the same value
 eg == operator
- What is N-gram?
 is a continuos sequence of n items(words,symbols,letters)from a given sample
 n-gram models are used in nlp
 to predict the next item in sequence based
 useful in speech recognition,machine translation,spelling correction
- What are Limitations of LLM?
 LLM - limitations of large language models
 lack of true reasoning
 bias and fairness
 black box problem

computational intensity
prone to factual errors

- What is sentiment Analysis?

is an nlp technique used to determine whether a piece of text is positive, negative, neutral

it involves identifying, extracting information

- What is State-Space Search?

is a problem solving approach in ai where the problem is represented as a set of possible states and the goal is to find a path from an initial state to goal state

is represented as a tree or graph

- State any two approaches to knowledge representation.

logical representation - uses logic (like propositional or first order logic) to store knowledge in a clear manner. eg using if then stmts

semantic network - represents knowledge in a graph format where nodes represent objects, entities and edges represent the relationship between them

- What are the applications of classical Planning?

robotics - planning the sequence of actions and paths

logistics and supply chain management - improve efficiency

automated scheduling - allocating resources

space missions - developing satellites

game AI -

problem solving

- Differentiate between propositional logic and first-order logic? Write the State less desirable properties of knowledge.

propositional logic - propositions (atomic stmts)

limited true or false stmts

eg. p(it is raining)

first order logic - predict constants, variables

represent relationships

eats(x,y)(everyone eats anything)

properties :

complexity : developing effective knowledge representation

scalability:

lack of common sense

inaccuracies

ambiguity

- Explain Hierarchical Planning in Autonomous Driving.

hierarchical planning breaks down the complex task of autonomous driving into a series of smaller

high level plan

intermediate level plan

low level plan

- What is Neural Language Models?

a nlm is a type of language model that uses artificial neural networks to learn the statistical properties

language modelling task

neural networks

distributed representation

contextual understanding

machine translation

text generation

speech recognition

sentiment analysis
autocorrect

- What is an Information Retrieval(IR) Model?
an IR is a mathematical framework or a set of processes that defines how documents and queries are represent
retrive info from large collection
eg.boolean model
- What are the Application of Reinforcement Learning in Robotics?
learning locomotion - teaching robots to walk,run
text execution - training robots to perform task
human-robot interaction - robots interact humans
- Define constraint satisfaction problem. Solve SEND+MORE=MONEY Using
constraint satisfaction problem
csp are mathematical questins defined as a set of objects whose state
must statisfy a no.of constraints or limitations
solving SEND+MORE=MONEY using csp
variables : S,E,N,D,M,O,R,Y
domains : {0,1,2,3,4,5,6,7,8,9,} for each variable
constraits:
1.all lettrs must have unique digit
2.S and M cannot be 0
3.the arithmetic must hold true:
SEND+MORE=MONEY
we use backtracking search constraint
the unique solution is found :
S=9,E=5,N=6,D=7,M=1,O=0,R=8,Y=2
9567+1085=10652
- What is Goal-based planning?
GBP is an ai approach where an agent forms a sequence of actions to
achieve the goals
- Write any four applications of AI .
healthcare - xrays,mris
autonomous vehicles - self driving cars
customer service - chatboots
finance - detecting fraud transcation
- What is environment? Types of environment.
the external world that the agent interacts with
types :
fully observable vs.partially observable :
an environment is fully observable if the agent sensors give it aces to
the complete state of the environment at all times
static vs.dynamic :
a static environment does not change
a dynamic agent is acting
discrete vs.continuous :
the no. of possible states is finite in a discrete environment
in a continuous values are infinite
deterministic vs.stochastic :
deterministic environment is completely determined by the current state
and the agents action
in a stochastic environment there is an element of chance
- Write a Python program Accept the number and check it prime or not.
def prime(n):

```

        if n<=1 :
            return false
        for i in range(2,100):
            if n% i==0:
                return false
        return true
n=2
if prime(n)
    print("number is prime")
else
    print("number is not prime")

```

- What is difference between append () and extend () python methods?
 - append()-adds a single element to the end of the list
 - accepts a single element of any data type
 - increases the list length by exactly one
 - extend()-adds each element from an iterable to the end of the list
 - requires an iterable as an argument
 - increases the list length by the no.of elements in the iterable

argument